

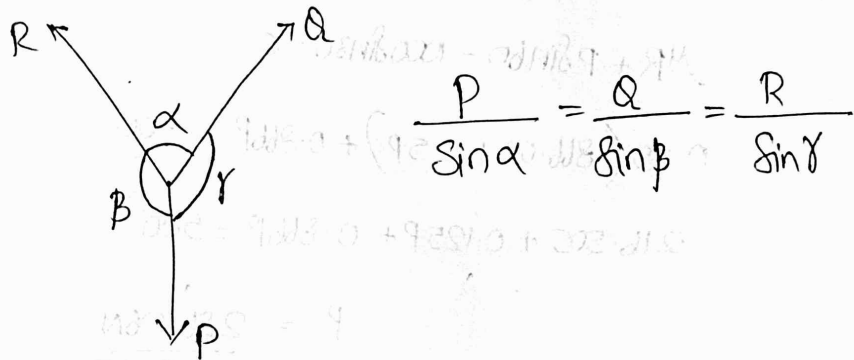
EST 100

ENGINEERING MECHANICS

DECEMBER 2020

1. Lami's theorem

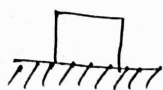
If three forces acting at a point are in equilibrium, each force will be proportional to the sine of angle between other two forces.



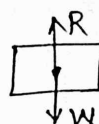
2. Free Body Diagram (FBD)

The diagram of the body in which the body under consideration is free from all contact surfaces and is shown with all the forces on it (including self weight, reactions from other contact surfaces) is called Free Body Diagram.

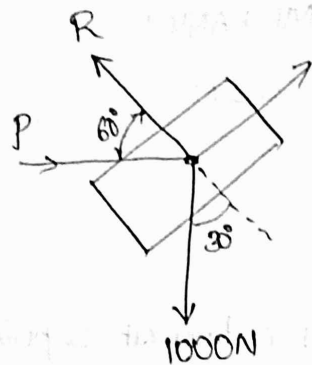
Examples: -



FBD of block



3.



ΣF Perpendicular to plane

$$R - 1000 \cos 30 - P \cos 60 = 0$$

$$R = 866.02 + 0.5P$$

ΣF parallel to plane = 0

$$F + P \sin 60 - 1000 \sin 30 = 0$$

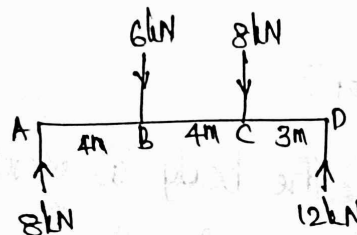
$$\mu R + P \sin 60 - 1000 \sin 30 = 0$$

$$0.25(866.02 + 0.5P) + 0.866P = 500$$

$$216.505 + 0.125P + 0.866P = 500$$

$$P = \underline{\underline{286.06 \text{ N}}}$$

4.



$$\Sigma F_y = 8 - 6 - 8 + 12 = 6 \text{ kN} \uparrow$$

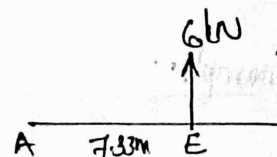
$$\Sigma M_A \Rightarrow 8(0) - (6 \times 4) - (8 \times 8) + (12 \times 11)$$

$$= -24 - 64 + 132 = 44 \text{ kNm anticlockwise}$$

By Varignon's theorem,

$$+(6 \times x) = +44$$

$$x = \frac{44}{6} = \underline{\underline{7.33 \text{ m}}}$$



5. $\vec{F} = 4\hat{i} + 4\hat{j} - 1\hat{k}$

$C(-2, 3, 5)$

$A(1, -2, 4)$

$\vec{M} = \vec{r} \times \vec{F}$

$\vec{r} = \vec{CA} = (1-2)\hat{i} + (-2-3)\hat{j} + (4-5)\hat{k}$

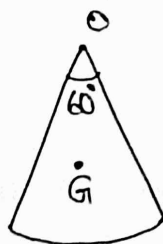
$\vec{r} = 3\hat{i} - 5\hat{j} - 1\hat{k}$

$\vec{M} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -5 & -1 \\ 4 & 4 & -1 \end{vmatrix}$

$= \hat{i}(5-4) - \hat{j}(-3-4) + \hat{k}(12-20)$

$= \underline{\underline{9\hat{i} - 7\hat{j} + 32\hat{k}}}$

6.



$r = 220 \text{ mm}$

$\alpha = 30^\circ$

$OG = \frac{2r \sin \alpha}{3\alpha} = \frac{2 \times 220 \times \sin 30}{3 \times \frac{\pi}{180} \times 30} = \underline{\underline{140 \text{ mm}}}$

7. $m = 50 \text{ kg}$

$u = 10 \text{ m/s}$

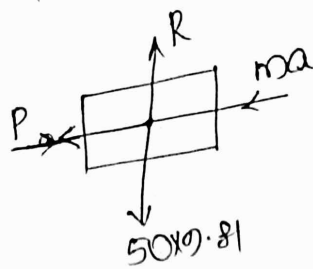
$v = 0$

$t = 5 \text{ sec}$

$v = u + at$

$0 = 10 + a \times 5$

$a = -10/5 = -2 \text{ m/s}^2$



$$-P - ma = 0$$

$$P = -ma$$

$$= 50 \times 2$$

$$P = \underline{\underline{100N}}$$

$$8. \quad R = 3H$$

$$R = \frac{u^2 \sin 2\theta}{g}, \quad H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\frac{u^2 \sin \theta \cdot \cos \theta}{g} = 3 \times \frac{u^2 \sin \theta \cdot \sin \theta}{2g}$$

$$2 \cos \theta = \frac{3}{2} \sin \theta$$

$$\frac{4}{3} = \tan \theta$$

$$\theta = \tan^{-1}(4/3) = \underline{\underline{53.13^\circ}}$$



$$9. \quad u = 40 \text{ kmph} = 11.11 \text{ m/s}$$

$$v = 50 \text{ kmph} = 13.88 \text{ m/s}$$

$$s = 300 \text{ m}$$

$$v^2 = u^2 + 2as$$

$$13.88^2 = 11.11^2 + 2 \times a \times 300$$

$$a = 0.1157 \text{ m/s}^2$$

$$d = 1\text{m}$$

$$r = 0.5\text{m}$$

$$a = r\alpha$$

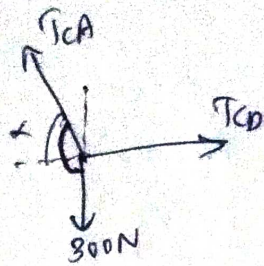
$$\alpha = \frac{a}{r} = \frac{0.1157}{0.5} = \underline{\underline{0.2315 \text{ rad/s}^2}}$$

10. Curvilinear motion

Curvilinear motion is defined as motion that occurs when a particle travels along a curved path.

Projectile motion

Projectile motion is a specific type of curvilinear motion. The motion of projectile may be thought of as the result of two separate, simultaneously occurring components of motions. One component along horizontal direction without any acceleration and other along vertical direction with constant acceleration due to force of gravity.

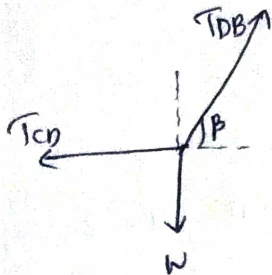


$$\frac{T_{CA}}{\sin 90} = \frac{T_{CD}}{\sin (90 + \alpha)} = \frac{300}{\sin (180 - \alpha)}$$

$$\frac{T_{CA}}{1} = \frac{T_{CD}}{0.845} = \frac{300}{0.484}$$

$$T_{CA} = 619.78 \text{ N}$$

$$T_{CD} = \underline{\underline{542.3 \text{ N}}}$$

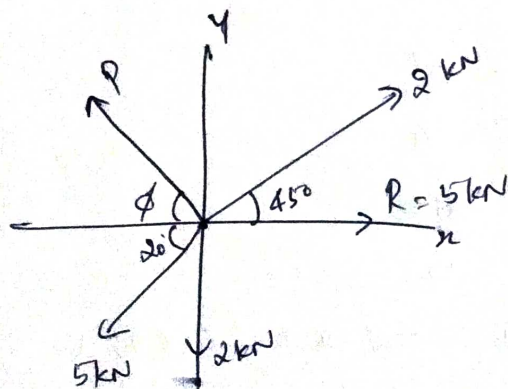


$$\frac{W}{\sin (180 - \beta)} = \frac{T_{CD}}{\sin (90 + \beta)} = \frac{T_{DB}}{\sin 90}$$

$$W = \frac{542.3 \times \sin (180 - 46.56)}{\sin (90 + 46.56)}$$

$$W = \frac{393.76}{0.688} = \underline{\underline{572.66 \text{ N}}}$$

11 (b)



$$\sum F_y = 0$$

$$\sum F_x = 5$$

$$\sum F_y = 2 \sin 45 - 2 - 5 \sin 20 + P \sin \phi = 0$$

$$P \sin \phi = 2.295 \quad \text{--- (1)}$$

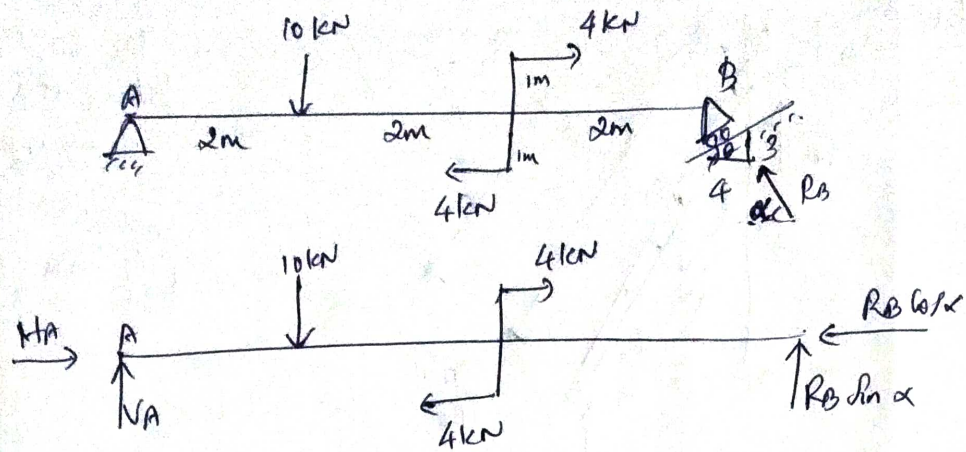
$$\sum F_x = 2 \cos 45 - 5 \cos 20 - P \cos \phi = 5$$

$$P \cos \phi = -8.28 \quad \text{--- (2)}$$

$$\frac{(1)}{(2)} \Rightarrow \tan \phi = 0.277, \quad \phi = 15.48^\circ$$

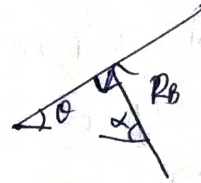
$$P = 8.11 \text{ kN}$$

(9)



$$\tan \theta = 3/4 \Rightarrow \theta = 36.86^\circ$$

$$\alpha = 90 - \theta = 53.13^\circ$$



$$\sum F_x = 0$$

$$H_A + 4 - 4 - R_B \cos \alpha = 0$$

$$H_A - R_B \cos 53.13 = 0 \quad \text{--- (1)}$$

$$\sum F_y = 0$$

$$V_A - 10 + R_B \sin \alpha = 0$$

$$V_A + R_B \sin 53.13 = 10 \quad \text{--- (2)}$$

$$\sum M_A = 0$$

$$(10 \times 2) + (4 \times 1) + (4 \times 1) - (R_B \sin \alpha) \times 6 = 0$$

$$(R_B \sin 53.13) \times 6 = 28$$

$$R_B = \underline{\underline{5.83 \text{ kN}}}$$

$$H_A = \underline{\underline{3.5 \text{ kN}}}$$

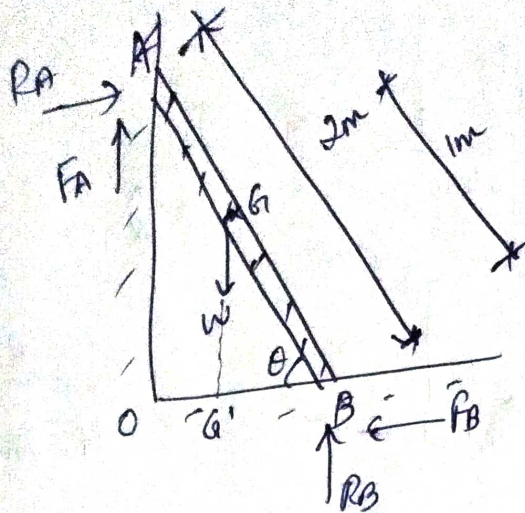
$$V_A = \underline{\underline{5.336 \text{ kN}}}$$

$$R_A = \sqrt{H_A^2 + V_A^2}$$

$$= \underline{\underline{6.38 \text{ kN}}}$$

$$\theta_A = \underline{\underline{56.738^\circ}}$$

13⑥



$$W = mg$$

$$W = 10 \times 9.81$$

$$= 98.1 \text{ N}$$

$$(\because m = 10 \text{ kg})$$

$$\Sigma F_x = 0$$

$$R_A - F_B = 0$$

$$R_A - 0.4 R_B = 0 \quad \text{--- ①}$$

$$\Sigma F_y = 0$$

$$R_B - 98.1 + F_A = 0$$

$$R_B - 98.1 + 0.6 R_A = 0$$

$$0.6 R_A + R_B = 98.1 \quad \text{--- ②}$$

$$\Sigma M_B = 0$$

$$R_A \times OA + F_A \times OB - W \times BG' = 0$$

$$R_A \times 2 \sin \theta + 0.6 R_A \times 2 \cos \theta - 98.1 \times 1 \cos \theta = 0$$

$$31.6 \times 2 \sin \theta + 0.6 \times 31.6 \times 2 \cos \theta - 98.1 \cos \theta = 0$$

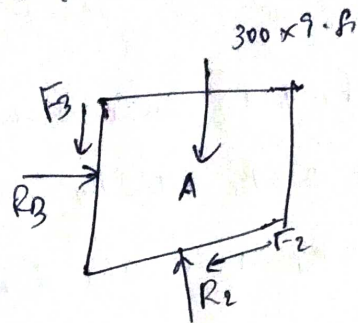
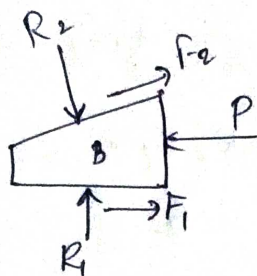
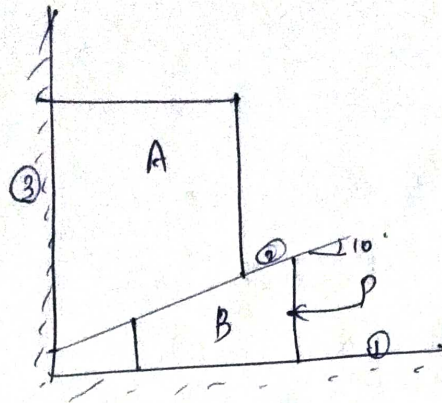
$$63.2 \sin \theta + 37.92 \cos \theta - 98.1 \cos \theta = 0$$

$$63.2 \sin \theta = 60.18 \cos \theta$$

$$\theta = \underline{\underline{43.59^\circ}}$$

$$R_A = 31.6 \text{ N}$$

$$R_B = 79.1 \text{ N}$$



Consider Block A FBD,

$\sum F_y = 0$

$$-(300 \times 9.8) - F_3 - F_2 \sin 10 + R_2 \sin 80 = 0$$

$$-0.3 R_3 - 0.3 R_2 \sin 10 + R_2 \sin 80 = 2943$$

$$0.933 R_2 - 0.3 R_3 = 2943 \quad \text{--- ①}$$

$\sum F_x = 0$

$$R_3 - F_2 \cos 10 - R_2 \cos 80 = 0$$

$$R_3 - 0.3 R_2 \cos 10 - R_2 \cos 80 = 0$$

$$-0.469 R_2 + R_3 = 0 \quad \text{--- ②}$$

$$R_2 = 3714.5 \text{ N}$$

$$R_3 = 1742.1 \text{ N}$$

Consider FBD of wedge

$$\Sigma F_y = 0$$

$$R_1 + R_2 \sin 10^\circ - R_2 \sin 80^\circ = 0$$

$$R_1 + 0.3 R_2 \sin 10^\circ - R_2 \sin 80^\circ = 0$$

$$R_1 - 0.933 R_2 = 0 \quad \text{--- (3)}$$

$$\Sigma F_x = 0$$

$$-P + R_1 + R_2 \sin 80^\circ \cos 10^\circ + R_2 \cos 80^\circ = 0$$

$$-P + 0.3 R_1 + 0.3 R_2 \cos 10^\circ + R_2 \cos 80^\circ = 0$$

$$0.3 R_1 + 0.469 R_2 = P \quad \text{--- (4)}$$

$$R_1 = 3465.6 \text{ N}$$

$$P = 2781.79 \text{ N}$$

$$P = \underline{\underline{2.78 \text{ kN}}}$$

15)

Sl No	fig	a_i	K_i	y_i	$a_i x_i$	$a_i y_i$
①		200×140 $= 28000$	$\frac{1}{3} \times 50$ $= \frac{200}{2} = 100$	70	2800000	1960000
②		$\frac{1}{2} \times 30 \times 80$ $= -1200$	$\frac{1}{3} \times 30$ $= 10$	113.33	-12000	-135996
③		$-\pi r^2 = -\pi \times 50^2$ $= -7853.98$	100	70	-785398	-549778.6
④		-40×60 $= -2400$	80	110	-432000	-264000

$$\bar{x} = \frac{\sum a_i x_i}{\sum a_i} = \frac{1570602}{16546.02} = \underline{\underline{94.92 \text{ mm}}}$$

$$\bar{y} = \frac{\sum a_i y_i}{\sum a_i} = \frac{1010225.4}{16546.02} = \underline{\underline{61.055 \text{ mm}}}$$

16) Magnitude of $P = 280 \text{ N}$, $Q = 260 \text{ N}$, $R = 210 \text{ N}$

Coordinates of $A = (0, 12, 0)$

$B = (-4, 0, -3)$

$C = (-4, 0, 6)$

$D = (6, 0, 4)$

$$\begin{aligned} \text{force vector } P &= 280 (\text{u.v along } \vec{AC}) \\ &= 280 \left(\frac{\vec{AC}}{|\vec{AC}|} \right) = 280 \left[\frac{-4\hat{i} - 12\hat{j} + 6\hat{k}}{\sqrt{(-4)^2 + (-12)^2 + 6^2}} \right] \end{aligned}$$

$$\begin{aligned} &= \frac{280}{17.42} (-4\hat{i} - 12\hat{j} + 6\hat{k}) \\ \vec{F}_P &= \frac{-1120\hat{i} - 3360\hat{j} + 1680\hat{k}}{17.42} \end{aligned}$$

force vector $Q = 260 (\text{u.v along } \vec{AB})$

$$= 260 \left(\frac{\vec{AB}}{|\vec{AB}|} \right)$$

$$= 260 \left[\frac{-4\hat{i} - 12\hat{j} - 3\hat{k}}{\sqrt{(-4)^2 + (-12)^2 + (-3)^2}} \right]$$

$$\begin{aligned} &= 20 (-4\hat{i} - 12\hat{j} - 3\hat{k}) \\ \vec{F}_Q &= -80\hat{i} - 240\hat{j} - 60\hat{k} \end{aligned}$$

$$\text{force vector } R = 210 \left[\text{u.v along } \vec{AD} \right]$$

$$= 210 \left[\frac{\vec{AD}}{|\vec{AD}|} \right]$$

$$= 210 \frac{[6i - 12j + 4k]}{\sqrt{6^2 + (-12)^2 + 4^2}}$$

$$= 210 \frac{[6i - 12j + 4k]}{14}$$

$$\vec{R} = 15 [6i - 12j + 4k]$$

$$\vec{R} = 90i - 180j + 60k$$

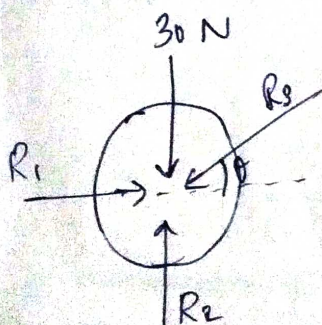
$$\text{Resultant force vector} = \vec{F}_P + \vec{F}_Q + \vec{F}_R$$

$$= (-80i - 240j + 120k) + (-80i - 240j - 60k) + (90i - 180j + 60k)$$

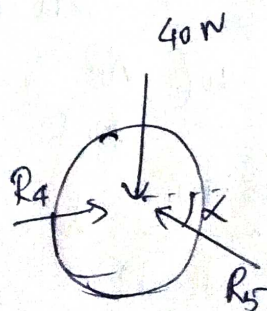
$$\vec{R} = -70i - 660j + 120k$$

$$\text{Resultant Magnitude} = \frac{674.46 \text{ N}}{|\vec{R}|}$$

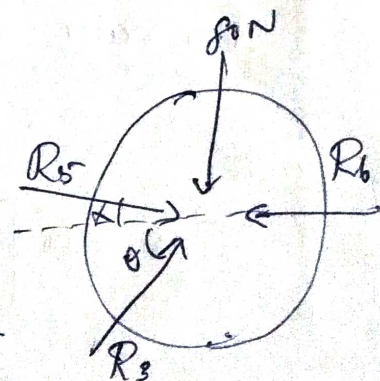
12)



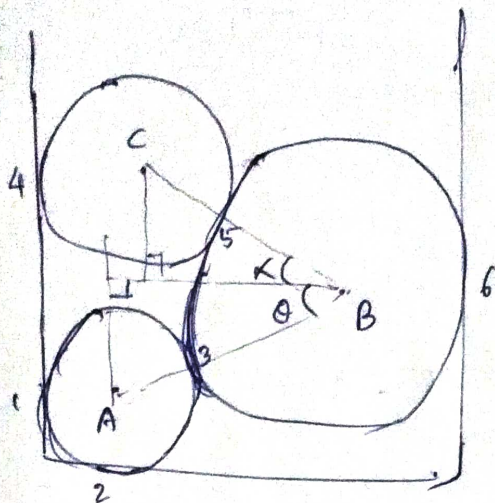
FBD of A



FBD of C



FBD of B



$$\cos \theta = \frac{360 - 86 - 120}{86 + 120}$$

$$\theta = 36.86^\circ$$

$$\cos \alpha = \frac{360 - 100 - 120}{100 + 120}$$

$$\alpha = 50.48^\circ$$

Consider FBD of C,

$$\sum F_x = 0$$

$$-R_5 \cos \alpha + R_4 = 0 \quad \text{--- (1)}$$

$$\sum F_y = 0$$

$$-40 + R_5 \sin \alpha = 0 \quad \text{--- (2)}$$

$$R_5 = 51.85 \text{ N}$$

$$R_4 = 32.99 \text{ N}$$

Consider FBD of B

$$\sum F_x = 0$$

$$-R_6 + R_3 \cos \theta + R_5 \cos \alpha = 0$$

$$R_3 \cos 36.86^\circ - R_6 = 32.99 \quad \text{--- (3)}$$

$$\sum F_y = 0$$

$$-80 + R_3 \sin \theta - R_5 \sin \alpha = 0$$

$$R_3 = \frac{R_5 \sin \alpha + 80}{\sin \theta} = 200 \text{ N}$$

$$R_6 = 127.03 \text{ N}$$

Consider FBD of A

$$\sum F_x = 0$$

$$R_1 - R_2 \cos \theta = 0$$

$$R_1 = 200 \cos 36.86^\circ$$

$$R_1 = 160 \text{ N}$$

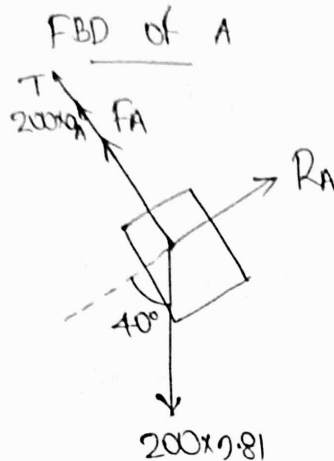
Reaction b/w cylinder A and vertical wall
~~160~~ $R_1 = 160 \text{ N}$

17.

$$M_A = 200 \text{ kg}$$

$$M_B = 100 \text{ kg}$$

$$\mu = 0.2$$



From the given figure,

$$Q_A = 2 Q_B$$

Block A slides down since

$$200 \sin 40 > 100$$

$$\sum F_{\perp \text{ to plane}} = 0$$

$$R_A - 200 \times 9.81 \cos 40 = 0$$

$$R_A = 1502.97 \text{ N}$$

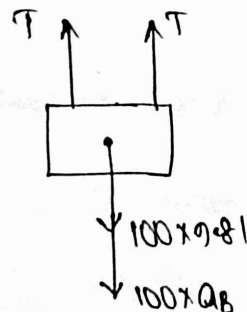
$$\sum F_{\parallel \text{ to plane}} = 0$$

$$T + F_A + 200 \times Q_A - 200 \times 9.81 \sin 40 = 0$$

$$T = 1261.14 - 200 Q_A - 0.2(1502.97)$$

$$T = 960.55 - 200 Q_A \quad \text{--- (1)}$$

FBD of B



$$2T - 100 \times 9.81 - 100 Q_B = 0$$

$$2T = 981 + 100 Q_B$$

$$T = 490.5 + 50 Q_B$$

$$T = 490.5 + 50 \times \frac{Q_A}{2}$$

$$T = 490.5 + 25 Q_A \quad \text{--- (2)}$$

By comparing (1) & (2)

$$\Rightarrow 960.55 - 200 Q_A = 490.5 + 25 Q_A$$

$$470.05 = 225 Q_A \Rightarrow Q_A = \underline{\underline{2.08 \text{ m/s}^2}}$$

Acceleration of block A = 2.08 m/s^2

Acceleration of block B = $\frac{a_A}{2} = \frac{2.08}{2} = \underline{\underline{1.04 \text{ m/s}^2}}$

From equation (2)

$$\Rightarrow T = 400.5 + 25a_A$$

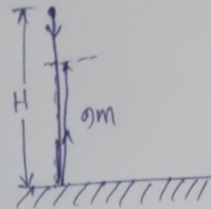
$$T = 400.5 + 25 \times 2.08$$

$$= \underline{\underline{542.5 \text{ N}}}$$

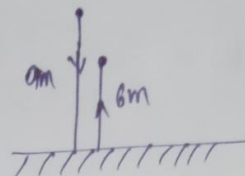


18.a.

Case 1



Case 2



From case 2,
Velocity of ball before striking floor,

$$u_1 = \sqrt{2gh_1} = \sqrt{2 \times 9.8 \times 9}$$

$$= \underline{13.28 \text{ m/s}}$$

Velocity of ball after striking floor,

$$v_1 = \sqrt{2gh_2} = \sqrt{2 \times 9.8 \times 6}$$

$$= \underline{10.84 \text{ m/s}}$$

There is no movement of floor, before
after striking, i.e. $u_2 = v_2 = 0$

Coefficient of restitution,

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$e = \frac{0 - (-10.84)}{13.28 - 0} = \underline{0.817}$$

Now in case 1,

Let velocity of ball before striking floor be u_1

velocity of ball after striking floor, $v_1 = \sqrt{2gh}$

$$v_1 = \sqrt{2 \times 9.8 \times 9} = \underline{13.28 \text{ m/s}}$$

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$0.817 = \frac{0 - (-13.28)}{u_1 - 0}$$

$$u_1 = \frac{13.28}{0.817} = \underline{\underline{16.25 \text{ m/s}}}$$

We know, $u_1 = \sqrt{2gH}$

$$16.25 = \sqrt{2 \times 9.81 \times H}$$

$$H = \frac{u_1^2}{2g} = \frac{(16.25)^2}{2 \times 9.81} = \underline{\underline{13.46 \text{ m}}}$$

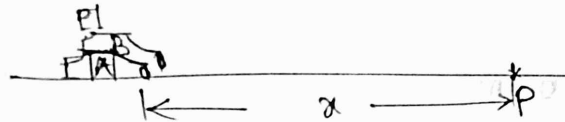
$$H = \underline{\underline{13.46 \text{ m}}}$$

18. b. $a_A = 0.75 \text{ m/s}^2$

$a_B = 1.1 \text{ m/s}^2$

$t_0 = 1.75 \text{ sec}$

$u_B, u_A = 0$



$s = ut + \frac{1}{2}at^2$

CAR A

$x = 0 + \frac{1}{2} \times 0.75 \times t^2$

$\Rightarrow \frac{0.75t^2}{2} = \frac{1.1(t-1.75)^2}{2}$

$= 0.75t^2 = 1.1(t^2 - 3.5t + 3.0625)$

$= 0.75t^2 = 1.1t^2 - 3.85t + 3.368$

$= 0.35t^2 - 3.85t + 3.36 = 0$

$\Rightarrow$ By solving this equation, $t = \underline{\underline{10.04 \text{ sec}}}$

Speed of car A at that time,

$v = u + at$

$= 0 + 0.75 \times 10.04$

$= \underline{\underline{7.53 \text{ m/s}}}$

Speed of car B at that time,

$v = u + at$

$= 0 + 1.1 \times (10.04 - 1.75) = \underline{\underline{9.10 \text{ m/s}}}$

Distance at which B overtakes A,

$$x = \frac{1}{2} \times 0.75 \times (10.04)^2$$

$$= \underline{\underline{37.80 \text{ m}}}$$

10. Q. $r = 30 \text{ cm} = 0.3 \text{ m}$

$$W_b = 145 \text{ N}$$

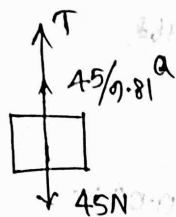
$$W_q = 45 \text{ N}$$

$$u = 0$$

$$h = 300 \text{ cm} = 3 \text{ m}$$

$$I = \frac{mr^2}{2} = \frac{145/9.81 \times 0.3^2}{2} = 0.66 \text{ kgm}^2$$

FBD of Q



$$T = 45 - \frac{45a}{9.81}$$

We know,

$$I\alpha = T \cdot r$$

$$0.66 \cdot \alpha = \left(45 - \frac{45a}{9.81}\right) \cdot r$$

$$0.66 \alpha = \frac{45 \times 0.3}{9.81} - \frac{45 \times 0.3 \times 0.3}{9.81} \alpha$$

$$1.07$$

$$25 \alpha = 13.5 \Rightarrow \alpha = 12.61 \text{ rad/s}^2 \Rightarrow a = \underline{\underline{3.78 \text{ m/s}^2}}$$

$$\alpha = 5.4 \text{ rad/s}^2 \Rightarrow a = 0.345 \text{ m/s}^2 = 4.10 \text{ m/s}^2$$

$$S = ut + \frac{1}{2}at^2$$

$$3 = 0 \times t + \frac{1}{2} \times 3.78 \times t^2$$

$$t^2 = 1.587$$

$$t = \underline{\underline{1.26 \text{ sec}}}$$

b) Velocity v with it strikes the floor

$$v = u + at$$

$$v = 0 + 3.78 \times 1.26$$

$$= \underline{\underline{4.76 \text{ m/s}}}$$

20.

a. $W = 50 \text{ N}$

$$k = 8 \text{ N/cm} = 8 \text{ N}/10^{-2} \text{ m} = 8 \times 10^2 \text{ N/m}$$

$$T = 2\pi \sqrt{m/k}$$

$$= 2\pi \sqrt{\frac{50}{9.8 \times 8 \times 10^2}} = \underline{\underline{0.501 \text{ sec}}}$$

b. $x_1 = 10 \text{ cm} = 0.1 \text{ m}$

$$x_2 = 20 \text{ cm} = 0.2 \text{ m}$$

$$v_1 = 1.2 \text{ m/s}$$

$$v_2 = 0.8 \text{ m/s}$$

$$v = \omega \sqrt{r^2 - x^2}$$

$$1.2 = \omega \sqrt{r^2 - 0.1^2} \Rightarrow (1.2)^2 = \omega^2 (r^2 - 0.1^2) \text{ --- (1)}$$

$$0.8 = \omega \sqrt{r^2 - 0.2^2} \Rightarrow (0.8)^2 = \omega^2 (r^2 - 0.2^2) \text{ --- (2)}$$

$$\frac{(1)}{(2)} \Rightarrow \frac{1.44}{0.64} = \frac{r^2 - 0.01}{r^2 - 0.04}$$

$$1.44 (r^2 - 0.04) = 0.64 (r^2 - 0.01)$$

$$1.44 r^2 - 0.0576 = 0.64 r^2 - 0.0064$$

$$0.8 r^2 = 0.0512$$

$$r^2 = 0.064$$

$$r = \underline{\underline{0.252m}}$$

From eq. (1)

$$\Rightarrow 1.44 = \omega^2 (0.252^2 - 0.1^2)$$

$$1.44 = \omega^2 \times 0.053504$$

$$\omega^2 = 26.91$$

$$\omega = 5.187 \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \underline{\underline{1.21 \text{ sec}}}$$

$$V_{\text{max}} = \omega \cdot r$$

$$= 5.187 \times 0.252 = \underline{\underline{1.307 \text{ m/s}}}$$

$$a_{\text{max}} = \omega^2 r$$

$$= (5.187)^2 \times 0.252 = \underline{\underline{6.781 \text{ m/s}^2}}$$